



Using HFSS and Linked BCs to Solve N-Fold Rotationally Symmetric Geometries

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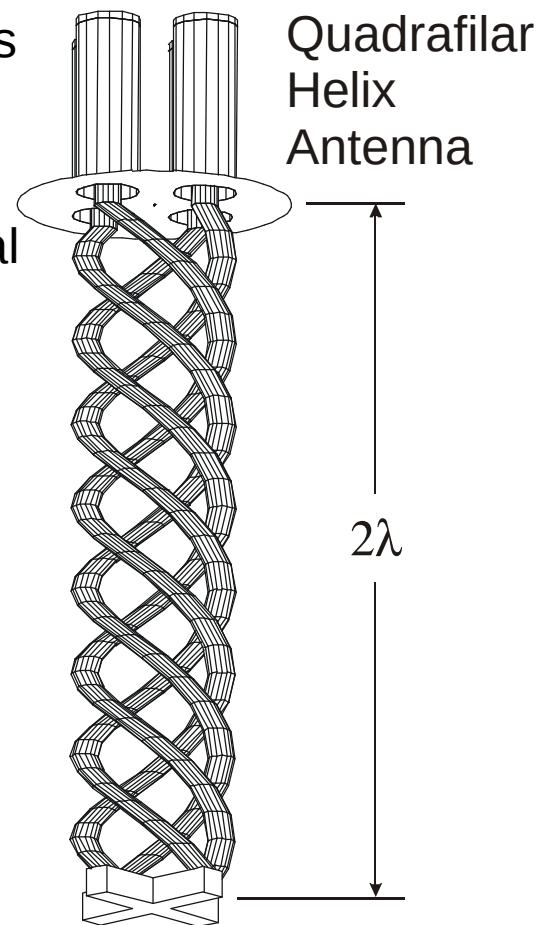
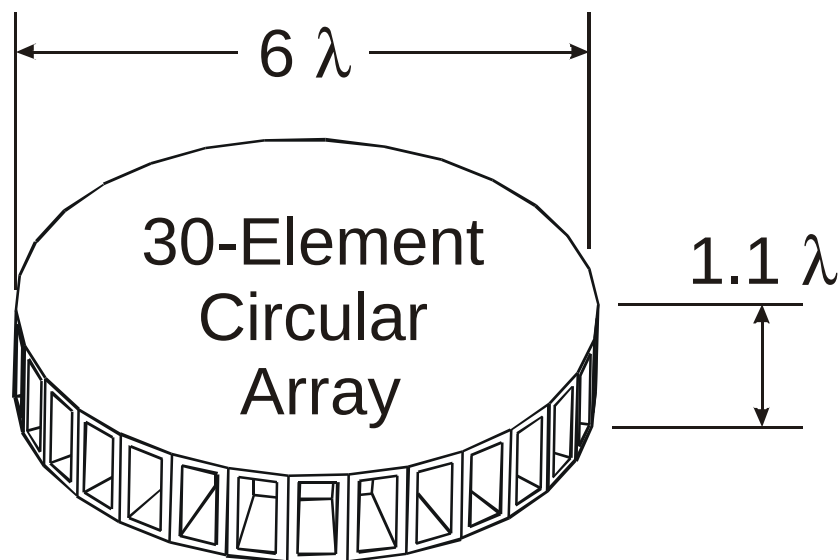
Introduction

- Presentation will show that linked boundary conditions (BCs) can be used to reduce problem size for geometries having N-fold rotational symmetry
- Necessary post-processing formulas will be given
- Significant CPU time savings will be demonstrated



Linked BCs Reduce Problem Size for Antennas Having N-Fold Rotational Symmetry

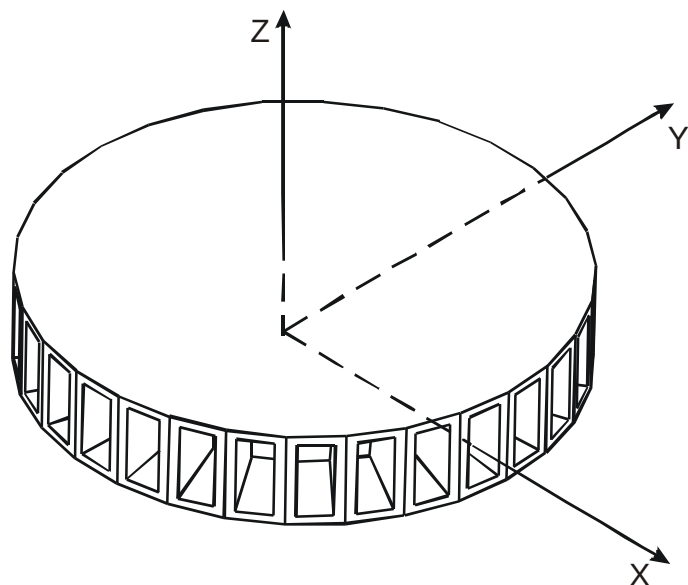
- CPU time and memory savings can be significant
- Presentation will illustrate with two different examples
 - 30-element circular array – does not have excitation symmetry
 - Quadrafililar helix antenna -- has both geometrical and excitation symmetry



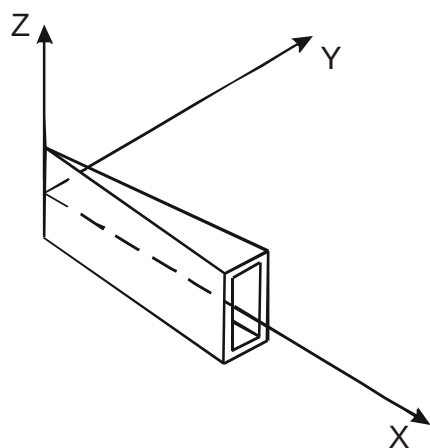


Solving 30-Element Circular Array

-- Full geometry vs Linked BCs



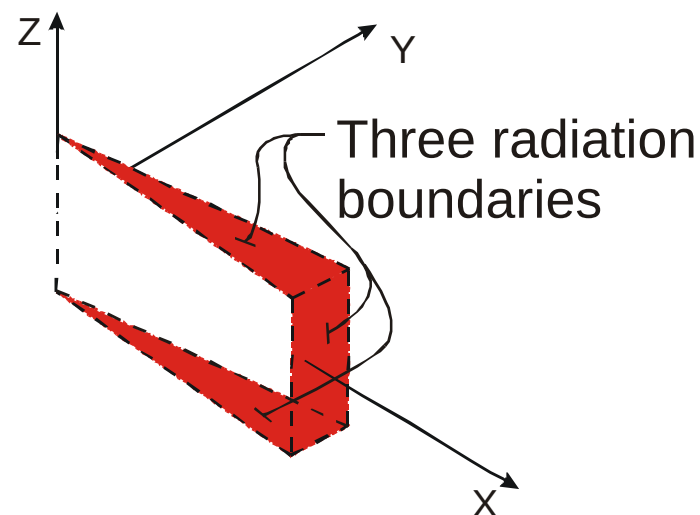
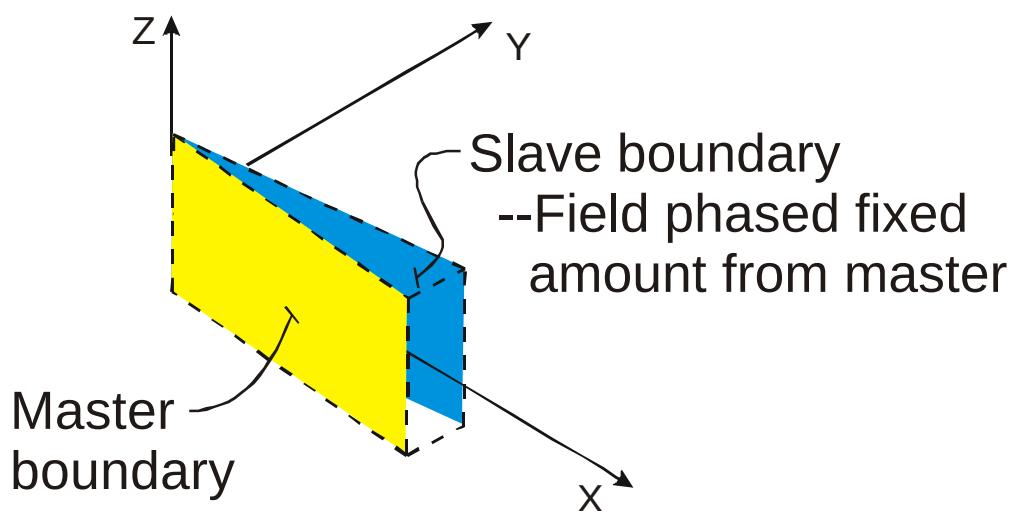
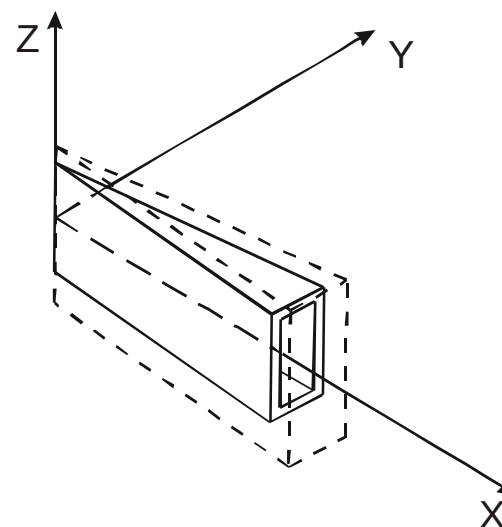
- Full geometry (without linked BCs)
 - Big volume
 - 30 ports
 - Solve project one time



- Linked BCs (12 degree unit cell)
 - Small volume
 - 1 port
 - Solve project 30 times
 - Ideal for parallel processing

Circular Array Requires Solving For 30 Different Linked BCs

- Solve with all possible conditions for phase of slave boundary (0, 12, 24, ..., 348 degrees)
- Post-process results to get entire array solution



Post-Processing Is Required To Get S-parameters and Radiation Patterns

- HFSS outputs active S-parameters
Passive S-parameters can be found using,

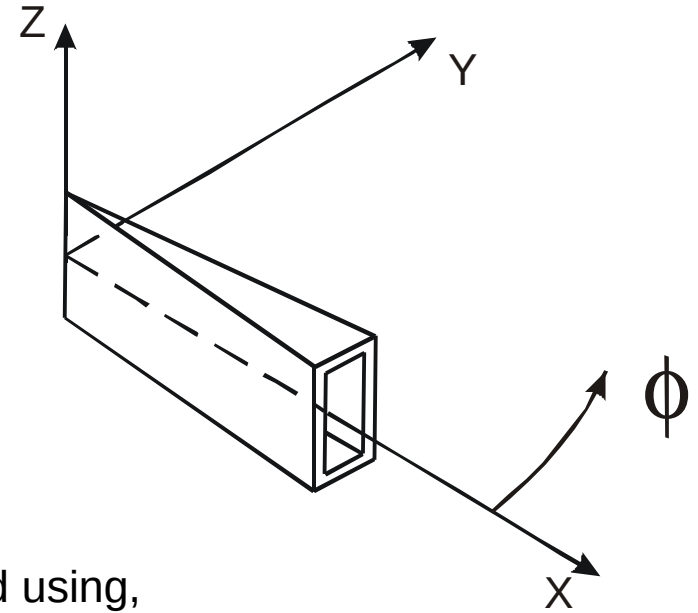
$$S_{lm} = \frac{1}{N} \sum_{i=1}^N (SA)_i e^{j2p(m-l)i/N}$$

where N = number of elements,
 $(SA)_i$ = active impedance for i^{th} mode

- HFSS outputs fields for a unit cell
 Embedded element pattern fields can be found using,

$$\overline{E} = \frac{1}{N} \sum_{i=0}^{N-1} \sum_{m=0}^{N-1} \overline{E}_i (f - 2mp/N) e^{j2pim/N}$$

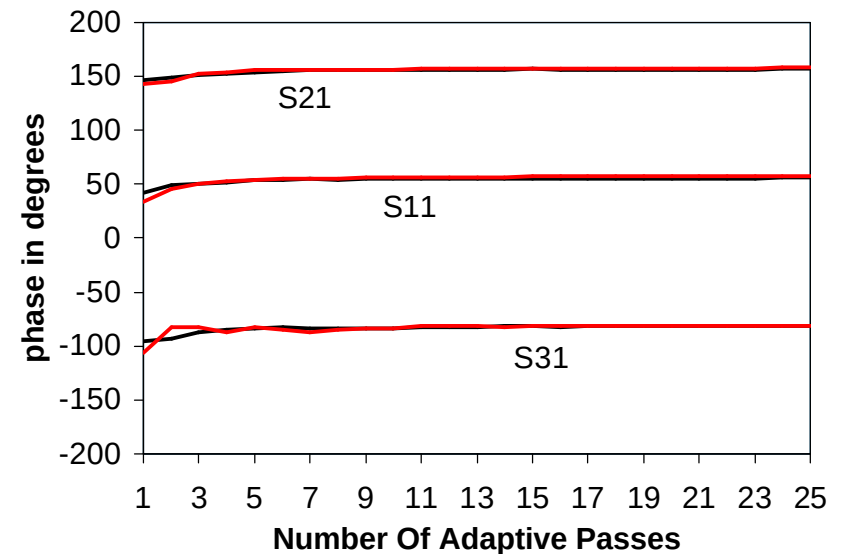
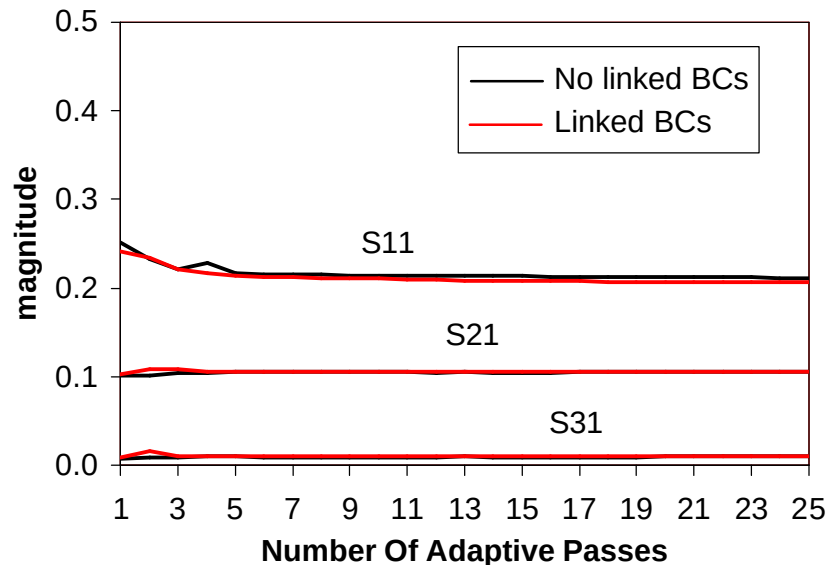
where $\overline{E}_i$ = field from i^{th} mode
 f = angle from x - axis in cylindrical coordinates





S-parameters Compare Closely For Solutions With And Without Linked BCs

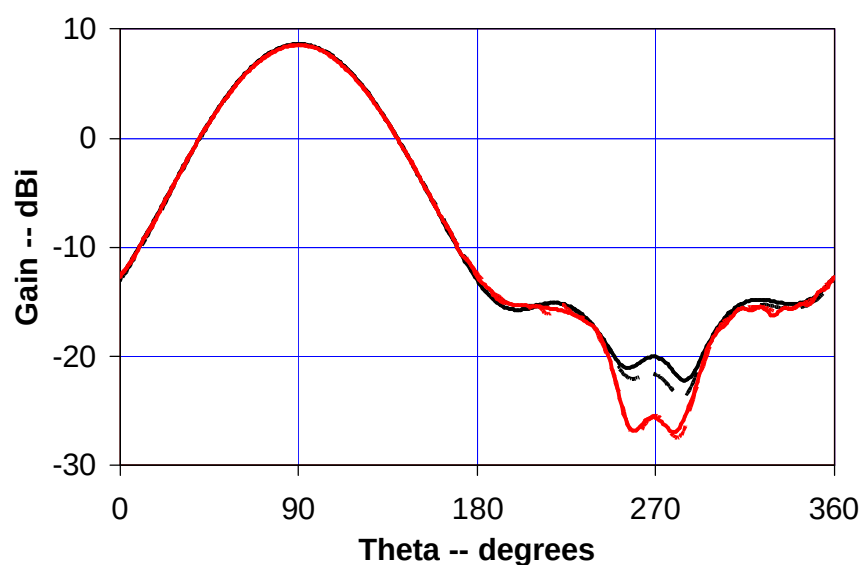
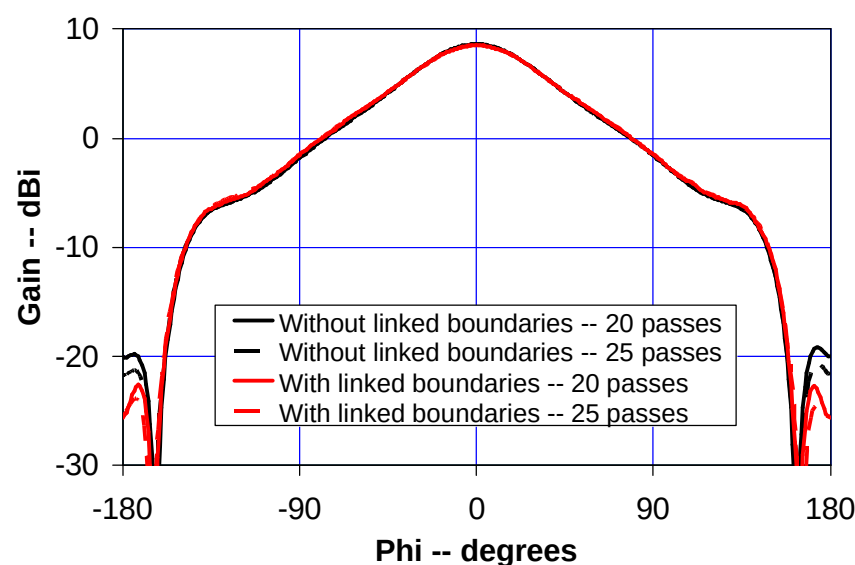
- Plots show S11, S21, and S31 versus number of adaptive passes
- Both solutions converge to same result within about 5 adaptive passes





Radiation Patterns Compare Closely For Solutions With And Without Linked BCs

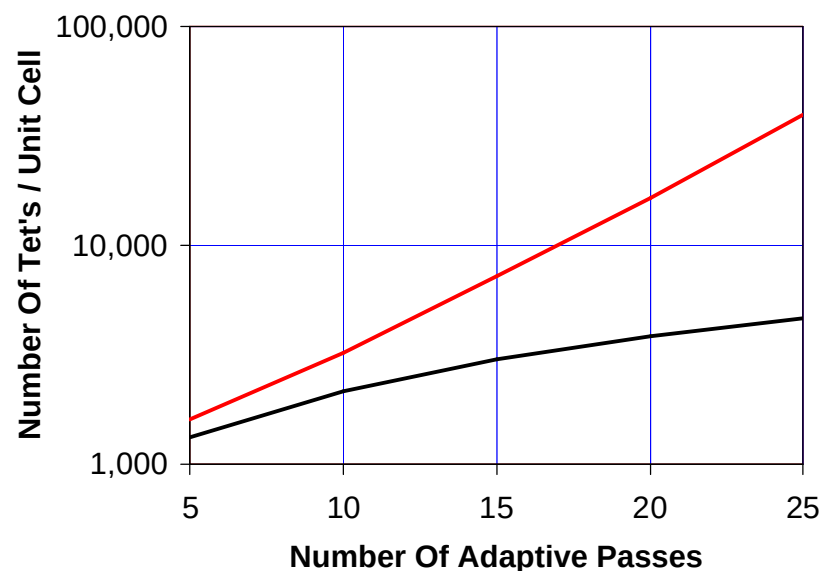
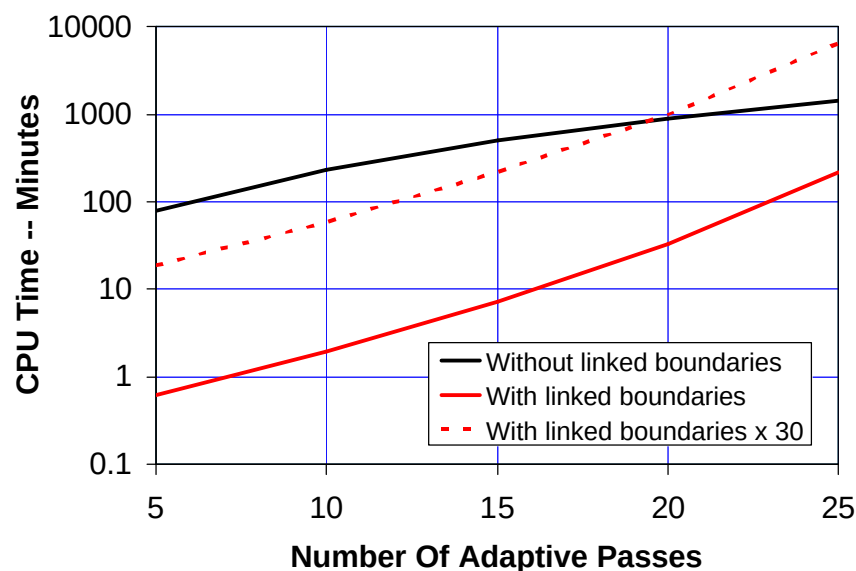
- Plots show embedded element pattern cuts in principal planes
- Only significant differences occur in backlobe region
- Appears that solution with linked boundaries is closer to convergence





When Compared Against # of Passes, Not Clear That Linked BCs Save CPU Time

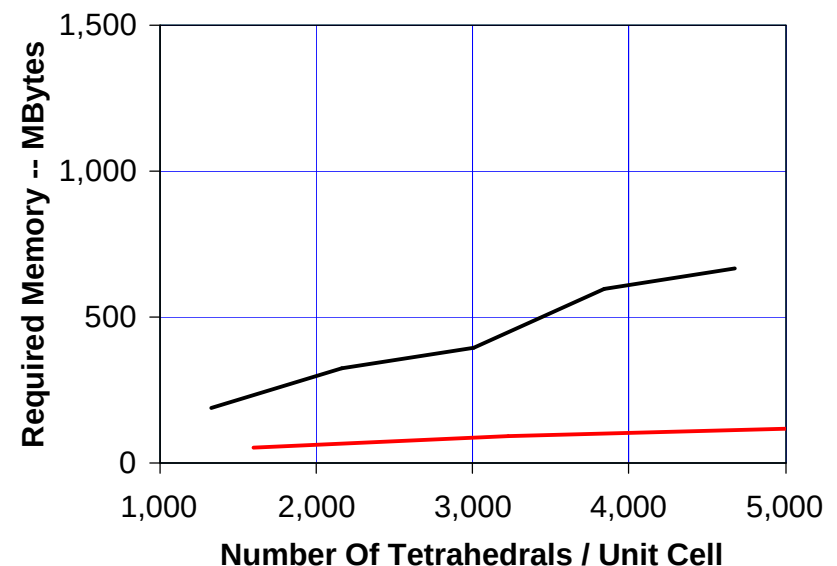
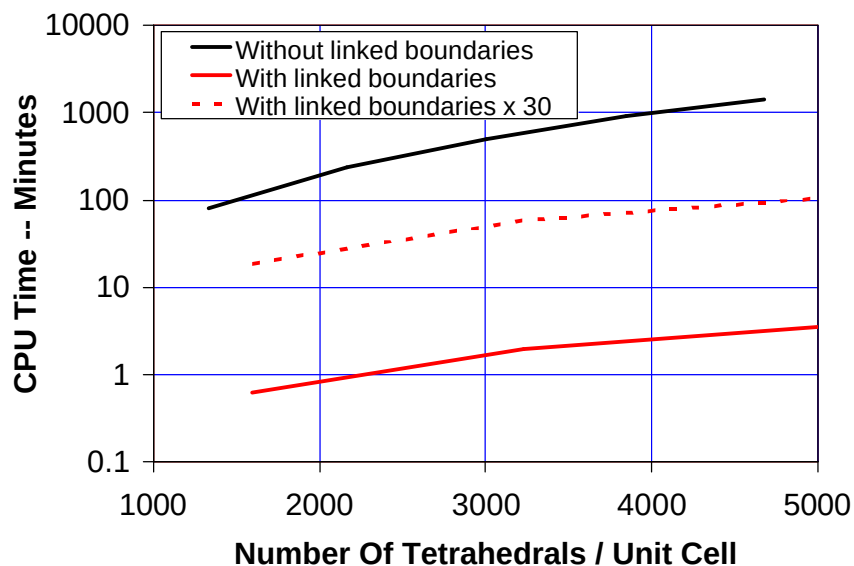
- Used 1.2 GHz processor with 1.5 GBytes of memory
- Plots show CPU time and # Tet's against number of adaptive passes
- Linked BCs solution faster only when # passes < 19





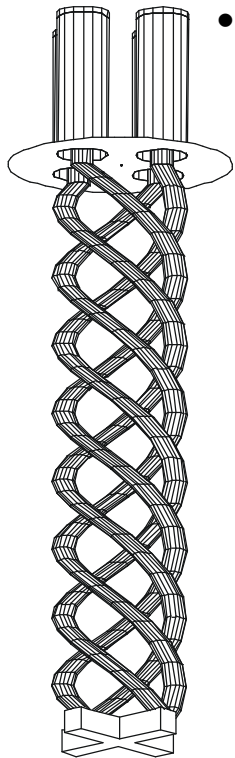
When Compared Against Number of Tet's per Unit Cell, Linked BCs Solution is Always Faster

- A better comparison of CPU times is vs number of tetrahedrals per unit cell
- Solution using linked boundaries is about ten times faster than solution without linked boundaries
- Solution using linked boundaries uses less memory

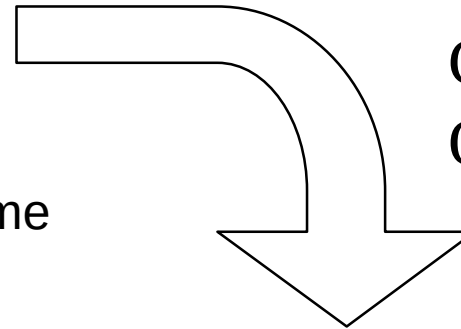


Solving Quadrafilar Helix Antenna

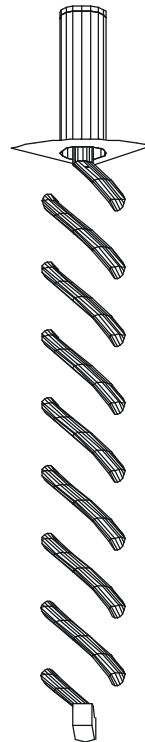
-- Full geometry vs Linked BCs



- Full geometry
 - Big volume
 - 4 ports
 - Solve project one time



CUT INTO
QUARTER



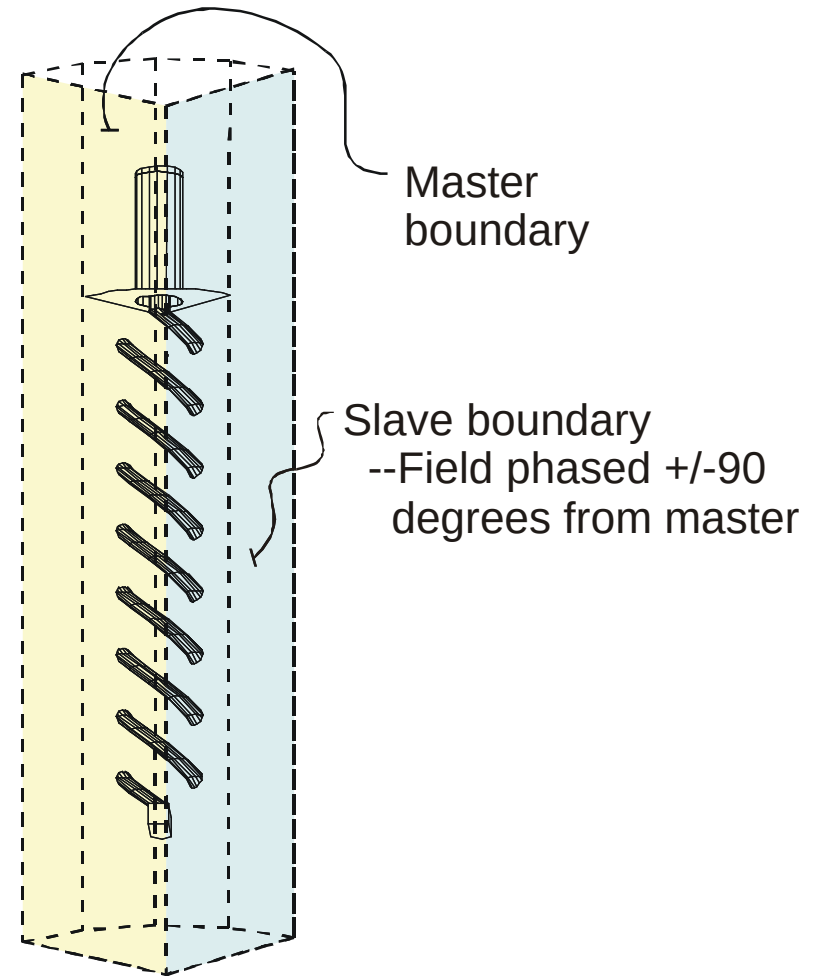
- Linked BCs (90 degree unit cell)
 - Smaller volume
 - 1 port
 - Solve project one time

Quadrafilar Helix Requires Solving Only One Linked BC

- HFSS outputs active S-parameters
- Radiation pattern found using,

$$\overline{E} = \sum_{m=0}^3 \overline{E}_1 (f - mp / 2) e^{jp m / 2}$$

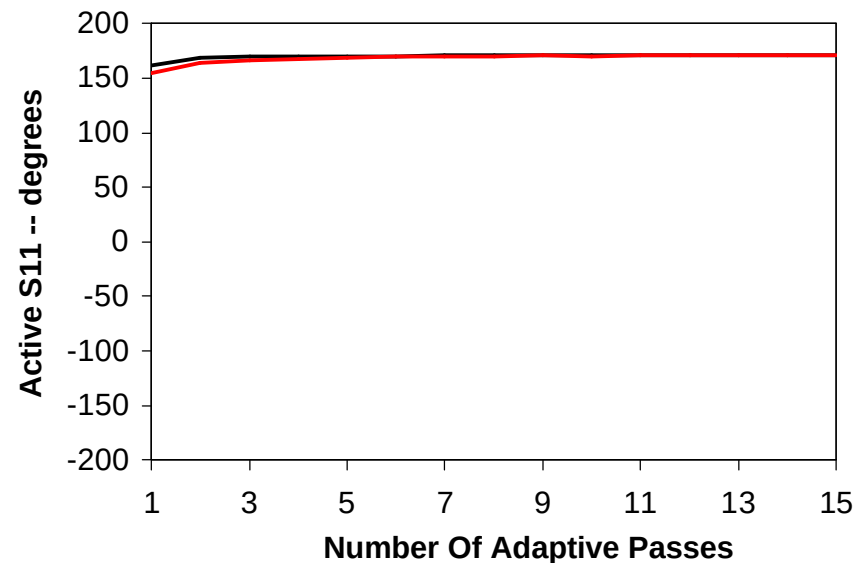
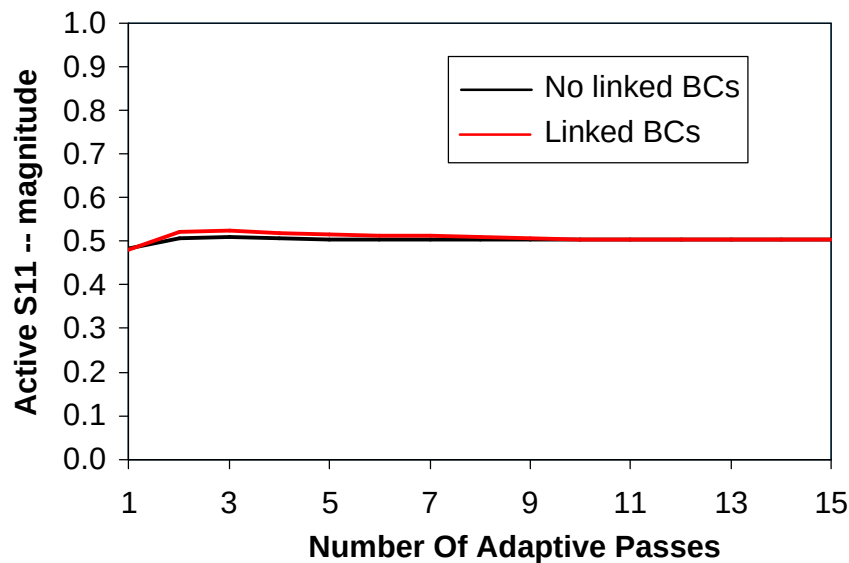
where $\overline{E}_1$ = field from desired mode (either RHCP or LHCP),





S-parameters Compare Closely For Solutions With and Without Linked BCs

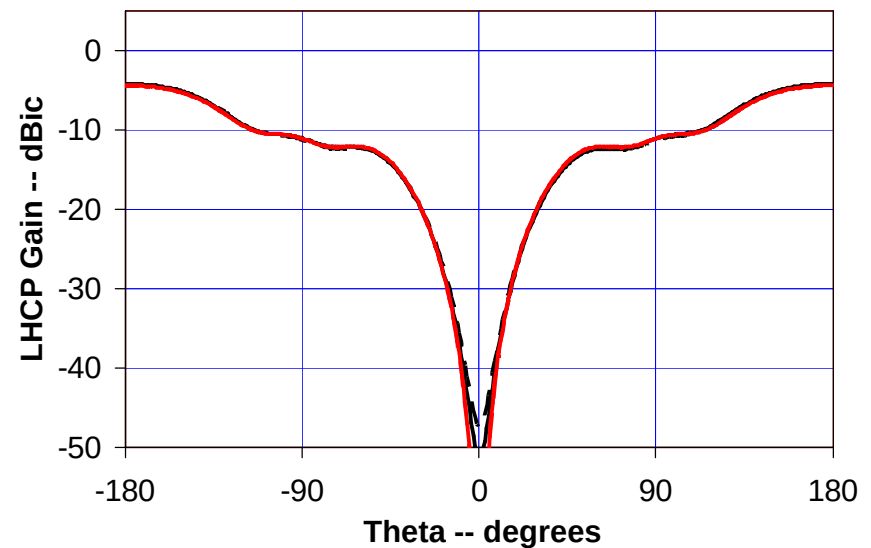
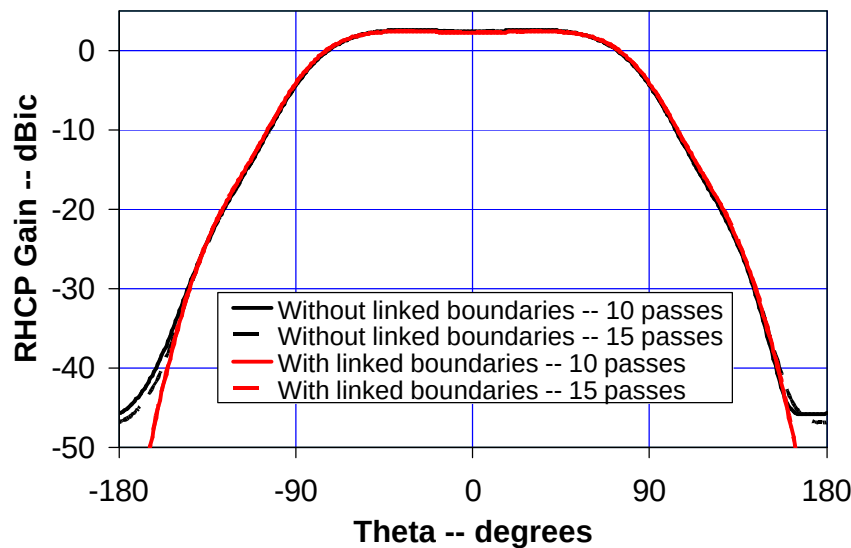
- Plots show active S11 versus number of adaptive passes
- Both solutions converge to same result within about five adaptive passes





Radiation Patterns Compare Closely For Solutions With and Without Linked BCs

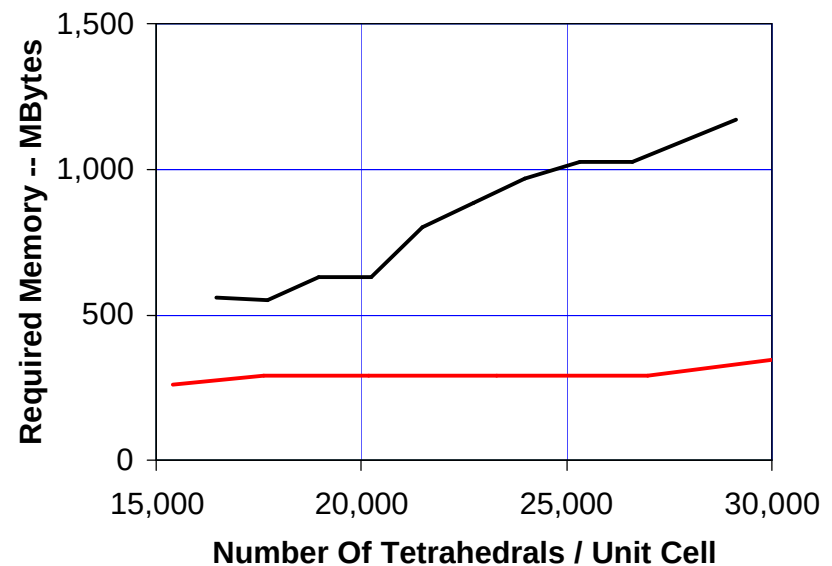
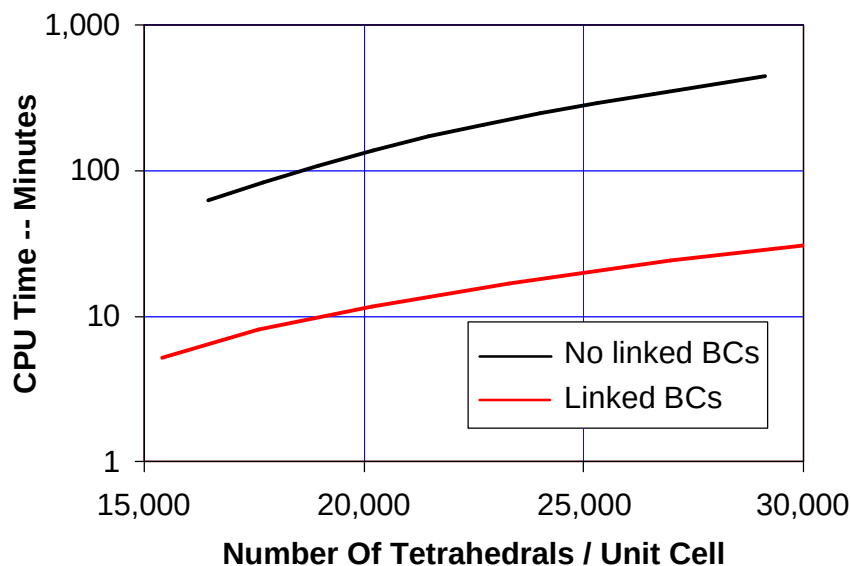
- Plots show RHCP and LHCP pattern cuts versus theta at phi = 0
- Only significant differences occur in backlobe region
- Appears that solution with linked boundaries is closer to convergence





CPU Time and Memory Usage Show Significant Advantage for Using Linked BCs

- Plots show CPU time and memory against number of Tet's per unit cell
- Solution using linked boundaries is about ten times faster than solution without linked boundaries
- Solution using linked boundaries uses less memory





Conclusions

- Demonstrated that linked BCs can reduce the required CPU time for antennas having N-fold rotational symmetry
- Post-processing is required to recover the radiation patterns from the HFSS output
- Largest time savings are for:
 - Problems where solving the entire geometry would exceed the computer memory
 - Problems, like the quadrafilar helix, that have n-fold symmetry in excitation, as well as in geometry